

Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: Diagnosing Class-Conditional Negative Transfer

Anonymous ACL submission

Abstract

Sarcasm detection requires intent beyond literal wording. LLM generation can expand sarcasm resources, but evaluation must distinguish sarcasm-relevant pragmatic cues from source-related cues. A shared encoder supports both sarcasm detection and machine-generated text detection (MGTD), yet source-predictive cues may distort sarcasm predictions. We harmonize Chinese and English resources and construct DualTest for source $\times$ sarcasm subgroup analysis, with prompt-specified style serving as the sarcasm label for AI-generated texts. On AI-generated texts prompted to be non-sarcastic, the final plain joint model’s false-positive rate exceeds that of its sarcasm-only counterpart by 67.0 percentage points, a pattern we call class-conditional negative transfer. Within AI-generated texts, AUC remains 0.82–0.88, showing preserved ranking, while score distributions shift upward. Both languages show the pattern, which persists when unchanged detectors evaluate a second test set generated by another LLM. Representation diagnostics link source-related scores and directions to the affected sarcasm decisions and show that removing one data-derived high-variance direction changes the subgroup gap. Exploratory analysis finds that predicted human-authorship probabilities do not consistently track LLM-judge ratings of human-likeness obtained without ground-truth source labels. These findings motivate source $\times$ sarcasm subgroup evaluation and external validation before detector probabilities are reused as quality scores or rewards.

1 Introduction

Sarcasm expresses an intended attitude that may differ from an utterance’s literal meaning, so automatic detection depends on implicit contrast, communicative intent, and context. Errors can reverse sentiment predictions and disrupt opinion-mining or content-moderation systems (Tay et al., 2018;

Vidgen et al., 2019). Annotation and model behavior also vary with elicitation method, annotator perspective, domain, style, and language (Oprea and Magdy, 2020; Casola et al., 2024; Jang and Frassinelli, 2024).

LLMs offer a way to expand sarcasm resources, but their usefulness depends on realizing the requested style without turning source-specific regularities into shortcuts. Evaluation should therefore separate source and sarcasm: both human-authored and AI-generated texts may be sarcastic or non-sarcastic.

A shared encoder with task-specific heads can reuse one representation for both predictions, but it also creates a route for interference: MGTD favors features that distinguish human-written from AI-generated text (Guo et al., 2023; Wang et al., 2024), whereas sarcasm detection should reflect intent rather than source. The labels themselves do not conflict; the risk lies in how the tasks organize the shared representation. If source-predictive cues influence the sarcasm head, information useful for MGTD but spurious for sarcasm can produce task conflict and negative transfer (Mueller et al., 2022; Hu et al., 2022b). This motivates testing how joint training changes sarcasm predictions.

Because source-predictive regularities and sarcasm cues vary across languages and corpora (Jang and Frassinelli, 2024; Casola et al., 2024; Wang et al., 2024), we examine the transfer pattern separately in Chinese and English.

DualTest supports source $\times$ sarcasm analysis, with prompt-specified style serving as the AI-text sarcasm label. Against same-configuration sarcasm-only controls, the joint procedure yields sharply higher false-positive rates on AI-generated texts prompted to be non-sarcastic. We call this class-conditional negative transfer because the degradation is concentrated in one source $\times$ sarcasm subgroup rather than distributed uniformly

across DualTest. The effect appears in both languages and persists when the same detectors evaluate outputs from another generator without retraining. Within AI-generated texts, sarcasm AUC remains 0.82–0.88, showing preserved ranking, while score distributions reveal an upward shift.

To connect decodable information with model behavior beyond probe accuracy (Hewitt and Liang, 2019; Elazar et al., 2021), we combine probes, task-head correlations, and representation interventions. Source-related scores and directions co-vary with the affected sarcasm decisions, and removing one data-derived high-variance direction changes the subgroup gap.

Detector scores can also enter generation rewards: ViSP includes a DIP-derived sarcasm score in its PPO reward (Wang et al., 2026; Wen et al., 2023). This motivates comparing detector sarcasm probabilities with LLM-judge sarcasm ratings. Separately, because predicted human-authorship probability does not measure perceived human-likeness, we compare those two signals as well, using ratings obtained without ground-truth source labels.

Contributions. We construct DualTest from Chinese and English resources and use same-configuration controls to identify class-conditional negative transfer concentrated on AI-generated texts prompted to be non-sarcastic. The effect persists across both languages and when the same detectors evaluate another generator’s outputs without retraining, while representation diagnostics link source-related scores and directions to the affected decisions and identify a high-variance direction for further causal analysis. An exploratory comparison with LLM-judge ratings further motivates external validation before detector probabilities are interpreted as measures of sarcasm quality or human-likeness.

2 Related Work

2.1 Sarcasm Detection, Synthetic Data, and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. Performance nevertheless varies substantially across datasets, which differ in label definitions, domains, languages, and modalities. Label schemes may reflect

intended or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024), while multimodal resources may use modality-specific annotations (Yue et al., 2024).

Recent studies evaluate LLM-generated sarcasm through predictive performance, diversity, linguistic properties, and human judgments (Majer et al., 2026), and associate synthetic-data characteristics with shortcut learning and poor transfer to human-authored text (Mahato et al., 2026). These findings motivate testing whether MGTD-derived source cues interfere with sarcasm detection in a shared encoder.

2.2 Machine-Generated Text Detection and Authorship Verification

MGTD includes likelihood-based zero-shot methods (Mitchell et al., 2023) and discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). M4 documents generalization gaps across generators, domains, and languages (Wang et al., 2024), whereas RAID stress-tests unseen generators, decoding variations, and attacks (Dugan et al., 2024). These findings are consistent with reliance on generator- or domain-specific source cues, which may also shape a shared representation used for sarcasm detection. Bevendorff et al. (2025) frame realistic LLM detection as open-set authorship verification rather than closed-set attribution; here, binary human-versus-AI classification serves as a controlled auxiliary task.

To the best of our knowledge, the interaction between sarcasm detection and MGTD under shared-encoder joint training has not been directly examined. Because source is the MGTD target but should not determine sarcasm, we test whether the shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, transfer depends on task relatedness and data-distribution alignment (Zhang et al., 2023; Wu et al., 2020), while gradient conflict can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Matched single- and multi-task comparisons show similar failures and suggest that models may infer task identity from input distributions (Mueller et al., 2022); other work analyzes functional specialization in multi-task Transformers (He and Choi, 2021; Li et al., 2023). A complementary shortcut-learning ac-

count is that models exploit training-predictive features that fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). Shared representations can strengthen such correlations (Hu et al., 2022b; Chai et al., 2025), while additional datasets can introduce them and reduce worst-group performance (Compton et al., 2023).

Mitigation approaches either isolate task-specific parameters (Pfeiffer et al., 2021, 2020; Karimi Mahabadi et al., 2021; Stickland and Murray, 2019) or modify information flow within a shared graph (Wu et al., 2019; Pilault et al., 2021). Our comparison of a plain shared encoder with two intermediate-representation variants follows the second direction and examines subgroup variation in negative transfer.

2.4 Detector Outputs as Evaluation and Reward Signals

Detector probabilities are sometimes reused as evaluation scores or generation rewards. For example, ViSP (Wang et al., 2026) includes a DIP-derived sarcasm score (Wen et al., 2023) in its PPO reward (Schulman et al., 2017). Such reuse assumes that a classifier’s continuous output tracks the intended quality dimension, an assumption not established by classification accuracy; optimizing a model-based proxy can also diverge from an external measure (Gao et al., 2023). As LLM judges are increasingly used to evaluate generated text (Zheng et al., 2023), we examine whether detector sarcasm probabilities agree with separately obtained LLM-judge sarcasm ratings. We also compare predicted human-authorship probabilities with judge-rated human-likeness because source classification and perceived human-likeness are related but distinct constructs.

3 Dataset Construction

Testing cross-task interference requires separate sarcasm and source labels, where source denotes human versus AI authorship rather than corpus. We pool the English iSarcasmEval data (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). Since none provides both labels, we map their annotations to a common binary sarcasm space, yielding 44,862 instances (approx-

mately 87% English). From these corpora, we construct SarcTrain for sarcasm supervision, AuthTrain for source supervision, DualTest for source $\times$ sarcasm evaluation, and GenTrain for generation fine-tuning (Appendix A).

SarcTrain. We reserve 2,103 eligible sarcastic records for GenTrain, then sample SarcTrain from the remaining 42,759 records. It contains 10,000 human-authored texts for sarcasm classification, balanced at 2,500 per language $\times$ label bucket. Three buckets are downsampled; the Chinese sarcastic bucket is expanded from 2,396 to 2,500 by sampling 104 records with replacement. We assign five folds after oversampling, so some duplicates cross folds (Appendix A).

AuthTrain. AuthTrain starts from 9,999 approximately language-balanced SarcTrain seeds (5,000 English; 4,999 Chinese) after seed-level quality control excludes one Chinese instance. Depending on the corpus, a seed is a topic, dialogue context, non-sarcastic rewrite input, or the text itself. The locally deployed Ollama model deepseek-r1:8b (Guo et al., 2025; Ollama, 2025) generates AI counterparts with four language $\times$ style templates, with style selected from each seed’s sarcasm label. Generation and quality control retain 9,996 counterparts (three unmatched seeds; 99.97% coverage), yielding 19,995 source-classification texts. Source and prompt-specified style are thus distinct, approximately balanced factors for source $\times$ sarcasm subgroup analysis (Appendix A).

DualTest. DualTest contains 4,664 texts for source $\times$ sarcasm evaluation. Its human subset comprises the official iSarcasmEval (1,400 English) and ToSarcasm (972 Chinese) test sets, which are excluded from the 44,862-record pool. Generation and quality control retain counterparts for 2,292 of 2,372 human texts (96.6% coverage). Human texts retain dataset labels; AI-text sarcasm labels record prompt-specified style rather than independent human ratings of perceived sarcasm. The four groups contain 823 human-sarcastic and 1,549 human-non-sarcastic texts, plus 779 AI texts prompted to be sarcastic and 1,513 prompted to be non-sarcastic. Record-level checks find no SarcTrain or AuthTrain overlap with either side of DualTest (Appendix A, which also gives the language $\times$ source $\times$ sarcasm distribution).

GenTrain. Separately, GenTrain contains 2,428 texts for the supervised fine-tuning (SFT) used in

§6. From 5,257 eligible sarcastic records, we sample 2,103 (40%; 1,214 English and 889 Chinese) and add 325 Chinese records sampled with replacement, yielding 1,214 instances per language. It contains 374 rewriting, 839 response, and 1,215 comment instances (Appendix A).

4 Joint and Single-Task Models

4.1 Tasks and Architecture

Given a Chinese or English input text x , the model predicts a binary sarcasm label and a binary source label, with all four label combinations treated as valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test how joint MGTD–sarcasm training changes the representation read by the sarcasm head. We compare three representation configurations and examine how subgroup errors vary across them. Figure 1 shows the RCNN configuration, whose intermediate module combines recurrent context with convolutional pooling for text classification (Lai et al., 2015), a design previously evaluated for multilingual sarcasm detection (Yang and Ikeda, 2024).

The RCNN configuration applies a single-layer BiLSTM followed by parallel one-dimensional convolutions and global max pooling. With hidden size 256 per direction, the BiLSTM yields a 512-dimensional vector at each token. We concatenate it with the 1,024-dimensional encoder output and apply kernels of widths 3, 4, and 5, each with 128 output channels. Global max pooling yields a 384-dimensional vector z , which passes through dropout to the two heads.

The plain configuration, used as the shared-encoder baseline, sends [CLS] directly to both heads. The second configuration uses a module termed task-branch interaction projection, which forms learned 256-dimensional sarcasm and authorship (MGTD) branch vectors h_s and h_a , and updates them as $h'_s = h_s + P_{a \rightarrow s}(h_a)$ and $h'_a = h_a + P_{s \rightarrow a}(h_s)$. Subscripts identify the branches, arrows give the mapping direction, and primes mark the updated vectors. Because each branch supplies a single key–value vector, all attention weights equal one, so the updates are residual cross-branch affine projections in which each P composes the corresponding value and output projections. We refer to the resulting full model as the interaction-projection configuration. All three configurations retain the same encoder and task definitions (Appendix B).

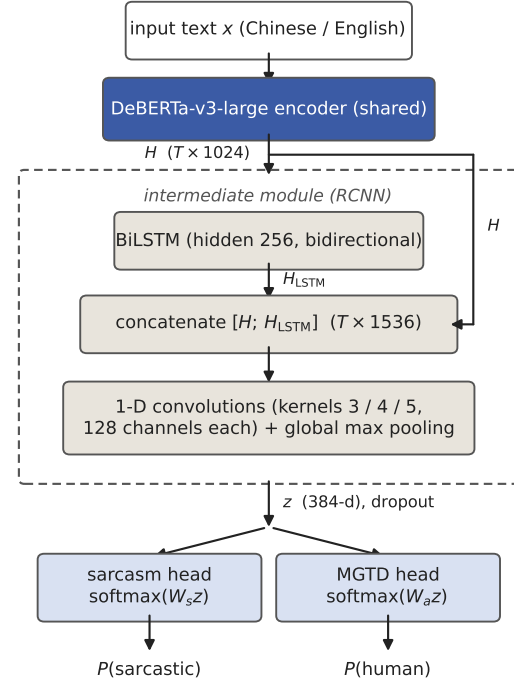


Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with two learned residual projections between the task branches, while retaining the same encoder backbone and task definitions (Appendix B).

4.2 Training Protocol

We distinguish representation configuration (plain, interaction-projection, or RCNN) from training procedure (joint, sarcasm-only, or MGTD-only): joint models have two task heads, whereas task-specific controls have one active head. We compare joint and sarcasm-only procedures in the plain and RCNN configurations. Within each pair, they share the encoder, representation configuration, sarcasm data, folds, and sarcasm-specific hyperparameters. The joint procedure also receives AuthTrain batches, updates shared parameters from both tasks, follows a mixed-task schedule, and uses joint early stopping. Compute exposure differs: the control traverses the full sarcasm fold each epoch, whereas the joint procedure samples sarcasm batches within the mixed schedule. The resulting contrast characterizes the complete joint-training procedure.

Training uses task-homogeneous batches in a 2:1 MGTD:sarcasm ratio, approximating the Au-

thTrain:SarcTrain size ratio. Let $z_s(x)$ and $z_a(x)$ be the representations read by the sarcasm and MGTD heads, and let W_s and W_a be their classifiers; subscripts s and a denote sarcasm and authorship (MGTD). The two representations coincide in the plain and RCNN configurations; in the interaction-projection configuration, they are the task-specific h'_s and h'_a defined above. For active task t , B_t is a task-homogeneous batch of input-label pairs (x, y) , and only its task loss is computed:

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(W_s z_s(x), y), & B_t \subset \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(W_a z_a(x), y), & B_t \subset \text{AuthTrain}, \end{cases} \quad (1)$$

where CE_w is sarcasm cross-entropy with class-weight vector $w = [1.0, 2.5]$ for the non-sarcastic and sarcastic classes, respectively. The weights were fixed before DualTest evaluation and are shared by the matched joint and sarcasm-only procedures. Cross-entropy is computed from logits, with softmax applied only for probability-based evaluation, and each task-homogeneous batch updates the active task head and the shared parameters. The encoder uses a lower learning rate than newly initialized modules. The sarcasm-head rate is fixed across configurations, and Appendix B reports the MGTD-head rates. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

For dual-task models, checkpoint selection accounts for both tasks. At epoch e , the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics. Single-task controls use their own validation loss with the same five-epoch patience (Appendix B).

We use five fold-matched runs. In run k , four SarcTrain folds train the sarcasm task and the fifth provides validation; all runs share a fixed AuthTrain 90/10 split, whose approximately 2,000-

example holdout provides a common MGTD validation set. Each run reloads its configuration’s pre-trained encoder and newly initializes the remaining modules and heads; all configurations use the same folds and seed schedule. DualTest is held out until evaluation. We report mean $\pm$ standard deviation across runs to summarize variation from the sarcasm fold and initialization; for subgroup rates from separately trained final models, Wilson intervals summarize uncertainty over DualTest examples (Table 3; Appendix C.8).

For each configuration used in the subgroup analysis, we train a final model on all applicable training data for a fixed epoch count determined from the five validation runs, using an independent seed and no early stopping. These models provide the subgroup and representation analyses but are excluded from the five-run aggregates; the exploratory analysis in §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

We compare backbones and joint- versus single-task procedures on DualTest using five-run aggregates, then use separately trained final models for source $\times$ sarcasm analysis, representation diagnostics, paired tests, and Wilson intervals.

5.1 Backbone Comparison

We compare dual-head mBERT (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), DeBERTa-v3-base, and DeBERTa-v3-large (He et al., 2023) under one protocol; for DeBERTa-v3-large, we also evaluate the interaction-projection and RCNN configurations. On DualTest, we report positive-class F1 for sarcastic texts and human-authored texts in the two tasks.

Under this protocol, the DeBERTa configurations yield higher mean F1 than mBERT and XLM-R, and RCNN achieves the highest means: 0.6309 for sarcasm and 0.9301 for MGTD (Table 1). These system-level comparisons jointly reflect backbone family, model scale, representation design, and optimization. On this empirical basis, the subsequent analyses use DeBERTa-v3-large, with the plain shared encoder as the reference and interaction-projection and RCNN as alternative representation pathways. Appendix C provides macro-averaged, human-subset, and language-stratified results.

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLm-RoBERTa-base	0.4603 $\pm$.057	0.8321 $\pm$.025
dual-head		
DeBERTa-v3-base	0.5646 $\pm$.056	0.8952 $\pm$.003
head		
DeBERTa-v3-large	0.6126 $\pm$.031	0.9221 $\pm$.010
dual-head		
+ interaction-projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Descriptive dual-head results across backbones and representation configurations (DualTest; five-run mean $\pm$ standard deviation; bold indicates the highest mean).

Config.	Evaluation task	Joint	Single-task	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	-6.08
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	+1.67
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	-4.43
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	+0.49

Table 2: Joint vs. corresponding single-task procedures (mean $\pm$ standard deviation over five fold-based runs; Δ = joint - single-task, in percentage points).

5.2 Joint versus Single-Task Procedures

To determine how performance differs between the joint and corresponding single-task procedures, Table 2 compares them for both the plain and RCNN configurations. Under the evaluated joint-training procedure, sarcasm F1 is lower in all five matched runs for both configurations, by 6.08 pp on average for plain and 4.43 pp for RCNN. Mean MGTD F1 is higher by 1.67 pp and 0.49 pp, respectively. The observed joint-single sarcasm gap is smaller for RCNN than for the plain configuration. We next use separately trained final models to characterize how the joint-single difference varies across source $\times$ sarcasm subgroups.

5.3 Source $\times$ Sarcasm Subgroup Analysis

We evaluate the four source $\times$ sarcasm subgroups separately. Each contains one sarcasm reference class, so we report recall for sarcastic groups and false-positive rate for non-sarcastic groups; human texts retain dataset annotations, whereas AI-text labels denote prompt-specified style.

5.3.1 Three Configurations and Sarcasm-Only Controls

Plain and RCNN provide same-configuration matched comparisons: within each pair, both models use the same all-human sarcasm data, while only the joint model also receives MGTD train-

ing. The interaction-projection configuration lacks a matched sarcasm-only control and is therefore descriptive. Table 3 reports the sarcasm predictions; corresponding MGTD results are in Appendix C.

On AI-generated texts prompted to be non-sarcastic, the plain joint and sarcasm-only false-positive rates are 86.3% and 19.3%, a 67.0-pp difference; the matched RCNN rates are 34.1% and 19.3%, a 14.8-pp difference. Exact paired McNemar tests give $p < .001$ for both comparisons (Appendix C.8).

Within AI-generated texts, sarcasm AUC is 0.82–0.88 across the models in Table 3, indicating that they retain the ability to rank sarcastic above non-sarcastic texts within this source. For the matched plain pair, the median sarcasm probability rises from 0.204 under sarcasm-only training to 0.767 under joint training; under joint training, the interquartile range is 0.635–0.839, entirely above the 0.5 decision threshold. These results show that joint training shifts sarcasm probabilities upward while leaving the ranking of AI-generated sarcastic and non-sarcastic texts largely intact.

The contrast between source groups is clearest for the plain pair: joint training increases the non-sarcastic false-positive rate by 4.3 pp on human-authored texts but by 67.0 pp on AI-generated texts.¹ In both language strata and both configurations, the joint model has a higher false-positive rate than its sarcasm-only counterpart on AI-generated non-sarcastic texts. For English ($n = 1,174$) and Chinese ($n = 339$), the differences are +63.5 and +79.4 pp, respectively, for plain, and +16.9 and +7.7 pp for RCNN (all paired McNemar $p < .001$). Thus, the negative transfer is most pronounced on AI-generated non-sarcastic texts, especially in the plain configuration, and recurs in both language strata.

5.3.2 Evaluation on Qwen3.5-Generated Texts without Detector Retraining

Using the same language $\times$ style templates, we evaluate the unchanged detectors on Qwen3.5-9B outputs (Ollama tag qwen3.5:9b) (Qwen Team, 2026). Without retraining, the false-positive rates on Qwen3.5-generated non-sarcastic texts are 91.3% and 7.9% for the plain joint and sarcasm-only models (gap 83.4 pp, versus 67.0 pp on DeepSeek), and 22.1% and 10.6% for the matched RCNN models. The higher joint-training rate

¹Differences use unrounded rates.

Model	H sarc. recall	H non-sarc. FPR	AI sarc. recall	AI non-sarc. FPR
Plain dual-head	67.1%	32.9%	98.2%	86.3% [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9% [38.5, 43.4]
RCNN dual-head	80.8%	40.5%	89.3%	34.1% [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3% [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3% [17.4, 21.4]

Table 3: Sarcasm predictions across four source $\times$ sarcasm groups. Human-authored texts use dataset sarcasm annotations; AI-generated texts use prompt-specified style labels. All dual-head rows use the joint procedure; single-head rows are sarcasm-only controls. Bold marks the focal AI non-sarcastic subgroup. Brackets show Wilson 95% intervals for the separately trained final models; full intervals and paired tests are in [Appendix C.8](#).

therefore persists on another generator, while magnitudes differ ([Appendix C.8](#)).

5.3.3 Representation-Level Diagnostics

Scope. We analyze frozen pooled representations from the final plain joint model underlying the 86.3% rate in [Table 3](#); the matched sarcasm-only model supplies the source-probe comparison. We test source decodability, alignment with sarcasm decisions, and whether fixed interventions change the AI–human false-positive-rate gap on non-sarcastic texts. For data-derived analyses, DualTest is stratified by the four source $\times$ sarcasm groups and divided equally into disjoint fitting and evaluation halves. The fitting half estimates intervention components; the evaluation half measures their fixed effects. All intervention results use this frozen final model.

Decodability and alignment. Five-fold logistic-regression source probes on the full DualTest representations give AUC 0.986 for joint and 0.974 for sarcasm-only representations, despite their sharply different subgroup gaps. Separately, on the full AI-generated non-sarcastic subset, the MGTD AI and sarcasm logits correlate at $\rho = 0.519$. Task-head directions are class-1 minus class-0 weights; the alignment analysis instead uses the normalized weight of a source probe fitted on the fitting half. The MGTD–sarcasm cosine is approximately -0.03 in Euclidean geometry. Covariance-aware alignment uses all pooled unlabeled DualTest representations and is transductive; its absolute value is 0.49 between task heads and 0.41 between the fitting-half source probe and sarcasm head. Together, these measurements characterize how source-related scores and directions co-vary with the affected sarcasm decisions.

Intervention protocol. The frozen model supplies the MGTD-head direction; the fitting half supplies probe, INLP, and PCA directions and

erasure transformations. These components are fixed and applied to the evaluation representations. Euclidean removal subtracts the centered projection while retaining the mean. The original sarcasm head measures subgroup false-positive rates. A separate source probe, cross-validated within each transformed evaluation half, measures residual linear source decodability. The unmodified evaluation-half rates are 84.8% for AI and 32.8% for human non-sarcastic texts (gap 52.0 pp), with source AUC 0.982.

Source-information erasure. LEACE ([Belrose et al., 2023](#)) estimates a rank-one covariance-aware erasure transformation from the fitting half. Applied unchanged to the evaluation half, it reduces source AUC to 0.589 and changes the AI and human false-positive rates to 68.7% and 49.5% (gap 19.1 pp). Their unweighted mean remains nearly constant (58.8% to 59.1%), showing error reallocation between source groups. Interventions removing the trained MGTD-head direction, the fitting-half source-probe direction, a Euclidean random direction, INLP subspaces of up to 16 directions ([Ravfogel et al., 2020](#)), or ten directions sampled in covariance-whitened space produce no comparable gap reduction.

High-variance sensitivity test. To construct srcORTH-PC1, we estimate a three-dimensional INLP source subspace on the fitting half by iteratively fitting, normalizing, and orthogonalizing three logistic-regression directions. We subtract the centered projection of the fitting representations onto this subspace and take the leading eigenvector of the residual covariance. Evaluation removes this direction alone. The AI and human false-positive rates change to 66.7% and 54.1% (gap 12.6 pp), while source AUC remains 0.982 at the reported precision. Error allocation is therefore sensitive to this high-variance direction in the orthogonal complement of the fitted INLP-3 sub-

space.

For this frozen model, removing the trained MGTD-head direction, the fitted source-probe direction, a Euclidean random direction, or an INLP subspace does not materially narrow the 52.0-pp source-group gap (resulting gaps: 51.9–72.1 pp). LEACE narrows the gap to 19.1 pp as source AUC falls from 0.982 to 0.589, whereas srcORTH-PC1 narrows it to 12.6 pp while source AUC remains 0.982 at the reported precision. Together, these interventions show that subgroup error allocation is sensitive to representation geometry beyond what the trained MGTD direction or residual linear source AUC alone captures, motivating semantic characterization and targeted causal tests (Appendix E).

6 Exploratory Comparison of Detector Scores with LLM-Judge Ratings

To test whether detector probabilities agree with independent ratings of generated text, we compare predicted human-authorship and sarcasm probabilities from RCNN with LLM-judge ratings of sarcasm and human-likeness from DeepSeek-V3 (DeepSeek-AI, 2024), Gemini-2.5-Flash-Lite (Google DeepMind, 2025), and GPT-5-nano (OpenAI, 2025) obtained without ground-truth source labels. We evaluate 600 paired outputs—one from base Qwen3-8B (Yang et al., 2025) and one from its QLoRA-adapted version fine-tuned on GenTrain (Dettmers et al., 2023)—for 300 prompts. Prompts comprise 180 Chinese topic-comment and 120 English rewriting tasks, so language and task are not independently varied.

In Chinese topic-comment generation, mean predicted sarcasm probability rises from 0.780 to 0.895 while the mean 1–5 LLM-judge rating falls from 3.99 to 3.80. Spearman ρ between probability and the mean available rating is -0.020 in Chinese and $+0.338$ in English. Predicted human-authorship probability rises in both settings while rated human-likeness declines, yielding a pooled ρ of -0.212 . However, the within-condition values of $+0.03$ for base outputs and -0.12 for SFT outputs show that opposing condition means dominate the pooled association.

The predicted-sarcasm change also differs between settings, increasing by 0.115 for Chinese topic-comment generation but decreasing by 0.124 for English rewriting. Across the six comparisons formed by two settings and three judges, fine-

tuned outputs receive lower mean sarcasm ratings in every case, and five differences remain significant after BH-FDR correction. Classification probabilities therefore do not consistently reproduce the ratings’ text-level rankings or condition-level directions.

Because pairwise judge rank correlations are low to moderate, we report judge-specific results and use the ratings for comparison rather than as a gold standard. Replacing the RCNN detector with the plain detector changes the corresponding correlations by at most 0.02. The resulting disagreements motivate external validation before detector probabilities are reused as quality scores or rewards. Appendix C provides the protocol and tests, and Appendix D presents selected cases.

7 Conclusion

We observe class-conditional negative transfer under the evaluated joint MGTD–sarcasm training procedure: on AI-generated texts prompted to be non-sarcastic, the final plain joint model’s FPR is 67.0 pp above that of its sarcasm-only counterpart; score distributions shift upward, while AUC within AI-generated texts remains 0.82–0.88. The higher joint-training FPR recurs in both languages and on Qwen3.5 outputs, with the RCNN and interaction-projection configurations showing lower observed false-positive rates on that subgroup. Representation interventions show that subgroup error allocation is sensitive to shared-representation geometry beyond what the trained MGTD direction or residual linear source AUC alone captures, motivating semantic characterization and targeted causal tests. The divergence between predicted human-authorship probabilities and human-likeness ratings motivates external validation before detector probabilities are used as quality scores or rewards.

Limitations

DualTest combines human sarcasm annotations with prompt-specified style labels for AI-generated texts. Accordingly, false positives for AI-generated texts are defined relative to the style specified in the prompt rather than independent human annotation of perceived sarcasm.

The core joint-versus-sarcasm-only comparisons evaluate the complete joint-training procedure. Within each pair, the architecture, sarcasm

data, fold assignments, sarcasm-head learning rate, and class weights are held fixed, while MGTD batches, mixed-task scheduling, shared-parameter updates, and joint early stopping are introduced together. Compute- and update-matched controls could help disentangle these components. Comparisons across architectures do not isolate architecture effects because model capacity and optimization also differ. The cost-sensitive sarcasm loss may also affect absolute recall and false-positive rates.

Because oversampling precedes fold assignment, some duplicated SarcTrain records occur in different folds and may appear on both sides of a training-validation split, introducing dependence into checkpoint selection, although these records do not occur in DualTest. Generation failures from deepseek-r1:8b leave 80 human test texts without retained AI counterparts.

The representation analyses on one separately trained final model connect source-related structure to behavior but do not isolate a unique causal feature or characterize variation across training seeds. The comparison between detector probabilities and LLM-judge ratings does not include human ratings, and language is not varied independently of generation task. Future work should test these findings using independent style annotations, matched training budgets, human ratings, and evaluation across a broader range of generators.

Ethical considerations

All human-authored texts were obtained from publicly released research datasets: iSarcasmEval, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. The iSarcasmEval and CSC repositories display MIT licenses, whereas we do not redistribute human-authored text from sources without an identified redistribution license (Appendix A). We collected no new user data or recruited participants, and we did not perform an additional audit of the released corpora for personally identifiable information or offensive content.

All AI-generated texts were produced locally using publicly released open-weight models (deepseek-r1:8b, Qwen3.5-9B, and the base and QLoRA-adapted Qwen3-8B). They were used only for training and evaluation and were labeled as AI-generated. For automated evaluation, evaluation inputs, generated outputs, and rat-

ing instructions were transmitted to the DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano APIs. Dataset metadata and ground-truth source labels were not transmitted.

Given the observed subgroup error and the absence of validation for individual-text authorship attribution or high-stakes deployment, detector outputs should not be used as evidence of authorship or misconduct without human review and domain-specific validation. They should not be deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The error concentrated on AI-generated texts prompted to be non-sarcastic is an example of systematic subgroup error affecting a particular type of generated content.

AI-assisted tools supported language editing, structural revision, and formatting, but did not generate the reported experimental outputs or statistics. The authors determined the study design, analyses, and conclusions and take full responsibility for the work.

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	A Detailed Dataset Construction,	1131
	Generation, and Quality Checks	1132
	A.1 Source corpora, sampling, and release	1133
	scope. The pooled collection contains 44,862 in-	1134
	stances in total (86.9% English and 13.1% Chi-	1135
	nese). SarcTrain is balanced across the four	1136

Corpus	Lang.	Inst.	Resource use
iSarcasmEval (SemEval-2022 T6)	English	3,467	SarcTrain; Gen- Train; DualTest*
CSC	English	6,878	SarcTrain; Gen- Train
ToSarcasm	Chinese	3,898	SarcTrain; Gen- Train; DualTest*
News Head- lines v2	English	28,619	SarcTrain
Open Chinese Internet Sar- casm Corpus	Chinese	2,000	SarcTrain

Table 4: Input corpora and resource use. The instance counts refer to the 44,862-record pooled collection; *DualTest uses separate official test splits excluded from these counts. Genres and annotations are iSarcasmEval (tweets, speaker self-annotation), CSC (conversational responses), ToSarcasm (topic comments, third-party annotation), News Headlines (distant supervision from The Onion / HuffPost), and the Open Chinese Corpus (Weibo, third-party annotation).

language $\times$ label groups by downsampling the two English groups and the Chinese non-sarcastic group, then sampling 104 additional Chinese sarcastic records with replacement. The Chinese sarcastic source pool contains 2,396 records corresponding to 2,394 distinct stored text values; after expansion to 2,500 records, its text-duplication rate is 4.24% (A.3).

The iSarcasmEval and CSC repositories display MIT licenses, whereas we did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm, News Headlines v2, or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from the latter sources. The planned release is limited to eligible AI-generated text, labels and metadata, and reconstruction scripts that require users to obtain the human-authored corpora from their original sources.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language $\times$ source $\times$ sarcasm distribution of DualTest. Human-authored texts use dataset sarcasm annotations; AI-generated texts use prompt-specified style labels.

A.2 Language $\times$ source $\times$ sarcasm distribution of DualTest.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Record-level matching finds that neither SarcTrain nor AuthTrain shares records with either the human or AI side of DualTest.

SarcTrain’s text-duplication rate is 0.34% for English and 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic group (4.24%; non-sarcastic group: 0.08%) and results primarily from the oversampling used to fill that group (A.1). Because oversampling precedes fold assignment, for 86 of the 104 added records, another copy originating from the same source record is assigned to a different fold. The two counting rules are summarized below.

Counting rule	Groups	Records	Cross-fold
Exact stored text; over-sampled copies included	116	241	98
Case/whitespace normalized; added copies excluded	19	46	16

The first rule underlies the reported duplication rates; the second isolates pre-existing duplicates after the added copies are removed. These relationships are relevant to fold-based checkpoint selection but do not place the affected training or validation records into DualTest. Post-processing detects and removes chain-of-thought-style markers (Wei et al., 2022) and the reasoning prefaces described in A.4.

A.4 AI generation protocol. The AI-generated texts in AuthTrain and the two DualTest versions are produced by locally deployed models through the Ollama API. The model deployed under the Ollama tag `deepseek-r1:8b` generates AuthTrain and the DeepSeek version of DualTest’s AI-generated subset. Qwen3.5-9B, deployed under the tag `qwen3.5:9b`, generates a second AI test subset that is evaluated with the same detectors without retraining. Both models use temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a maximum output length of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance’s sarcasm label. The templates constrain output style and length to 20–80 Chinese characters or 10–50 English words. These constraints

are intended to reduce, but do not eliminate, source differences in length, so [Appendix E](#) separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

```
【任务】生成一条讽刺风格的评论
【主题】{seed}
【要求】
- 直接给出简短的评论，不要解释思路或分析
- 风格要讽刺、幽默、夸张
- 表达与主题相关的观点即可，不必追求严谨
- 20-80 字为佳
【评论】
```

Its Hanyu Pinyin transliteration is:

```
[Renwu] Shengcheng yi tiao fengci
fengge de pinglun
[Zhuti] {seed}
[Yaoqiu]
- Zhijie geichu jianduan de pinglun;
buyao jieshi silu huo fenxi
- Fengge yao fengci, youmo, kuazhang
- Biaoda yu zhuti xiangguan de
guandian ji ke; bubi zhuiqiu yanjin
- 20--80 zi wei jia
[Pinglun]
```

Its English translation is:

```
[Task] Generate a sarcastic comment
[Topic] {seed}
[Requirements]
- Directly provide a short comment;
do not explain your reasoning or
analysis
- The comment should be sarcastic,
humorous, and exaggerated
- It is sufficient to express a
viewpoint related to the topic; the
comment need not be rigorous
- Preferably 20--80 Chinese
characters
[Comment]
```

The Chinese non-sarcastic template differs only in the style line, which requests a normal, natural style. The English templates use the same field structure and request either a sarcastic, humorous, exaggerated style or a normal, natural style, with a length limit of 10–50 words.

Post-processing removes `<think>...</think>` segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ń* ‘hmm’), “Okay,” “Let me,” and “First,” and Markdown markup. Outputs that still exceed the length limit or exhibit cyclic repetition after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-generated AI test subset has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use the full set of retained outputs for each generator rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from the `deepseek-r1:8b` outputs.

A.5 SFT training and generation evaluation.

After eligibility filtering, 5,257 records labeled as sarcastic remain for generation-task construction (2,972 English; 2,285 Chinese). After shuffling with seed 42, we sample 40% of these records, yielding 2,103 instances (1,214 English; 889 Chinese); language balancing then samples 325 additional Chinese records with replacement, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting instances, 839 response instances, and 1,215 comment instances. Generation evaluation uses a separately constructed set of 300 prompts that is not part of the GenTrain training-validation split (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. Language and generation task are therefore not independently varied. This set is used for detector scoring and all three LLM evaluations in §6.

B Hyperparameters and Training Details

Configuration	Enc.	Inter.	Sarc.	MGTD
Plain dual-head	5×10^{-7}	—	5×10^{-6}	1×10^{-5}
Plain	5×10^{-7}	—	5×10^{-6}	—
sarcasm-only				
Plain MGTD-only	5×10^{-7}	—	—	1×10^{-5}
Interaction-projection dual-head	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
RCNN dual-head	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
RCNN	5×10^{-7}	5×10^{-6}	5×10^{-6}	—
sarcasm-only				
RCNN	5×10^{-7}	5×10^{-6}	—	1×10^{-5}
MGTD-only				

Table 6: Discriminative learning rates for the dual-head and task-specific controls. The plain setting is also used in the backbone comparison. A dash denotes an inactive module or task head.

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5×10^{-6} in every configuration with an active sarcasm head. The MGTD-head rate is 1×10^{-5} for the plain dual-head model and the plain and RCNN MGTD-only controls, compared with 5×10^{-6} for the RCNN and interaction-projection dual-head models. Thus, the core joint-versus-sarcasm-only comparisons retain the same sarcasm-specific rates within each architecture. Because the RCNN joint-versus-MGTD-only comparison also changes the MGTD-head learning rate, it is reported descriptively rather than as an optimization-matched estimate of transfer; cross-configuration differences can likewise reflect optimization settings. Early stopping uses the normalized joint validation loss of Equation 2 for dual-head models and the task’s own validation loss for the task-specific controls, with a patience of five epochs in both procedures.

B.2 Shared training hyperparameters. Training uses `torch.optim.AdamW` with default PyTorch betas (0.9, 0.999) and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. We implement a linear 200-step warmup followed by a constant learning rate using PyTorch’s `LambdaLR`. Training runs for at most 10 epochs, with an early-stopping patience of 5 and gradient clipping at 1.0. The physical batch size is 4 with gradient accumulation over 4 steps, giving an effective batch size of 16 for the active task. The MGTD:sarcasm sampling ratio is 2:1, and the sarcasm class weights are [1.0, 2.5] for the non-sarcastic and sarcastic classes, respectively. The task ratio and class weights were fixed before DualTest evaluation. The normalized joint validation loss is $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{MGTD validation loss} / \text{first-epoch value}]$. The single-head controls process the full sarcasm training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

Detector inputs use a maximum sequence length of 512 tokens. In the RCNN module, a width- k convolution uses padding $\lfloor k/2 \rfloor$ followed by ReLU. Global max pooling covers all convolution positions; no post-convolution padding mask is applied.

All fold-based runs use a seed equal to 42 plus the fold index and share the same fold assignments and random-seed values across architectures and

ablations. The same fixed AuthTrain 90/10 split is reused in all five runs; only the SarcTrain validation fold changes. For each configuration, the epoch count used to train its final model is the mode of its five best epoch counts plus one, capped at 10 epochs; the resulting counts range from 3 to 10 across configurations. This mode-plus-one re-fitting rule was fixed before DualTest evaluation. The final model uses all applicable training data, a seed offset of +1000, no validation set, and no early stopping. Tables 1 and 2 report five-run means and standard deviations, whereas Table 3 and the subsequent fixed-model analyses use separately trained final models.

B.3 Model sizes and computational environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB; driver 570.124.04). Across the five-fold backbone, architecture, and ablation experiments and the training of final models, total compute is on the order of tens of GPU-hours. A single detector run takes from a few minutes to a few hours, and SFT takes about two hours. The software environment is Python 3.11.0, PyTorch 2.9.0 (CUDA 12.8; cuDNN 9.10.2), transformers 4.57.1, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0, pandas 2.3.3, peft 0.18.0, and datasets 4.4.1. AI-generated texts are produced locally through Ollama (Appendix A.4); LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.4 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Dettmers et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA configuration (Hu et al., 2022a) uses rank 64, alpha 16, and dropout 0.0 and targets `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses a learning rate of 5×10^{-5} , three epochs, batch size 1, gradient accumulation 8, and a warmup ratio of 0.05. After language balancing, a seed-42 split holds out 15% of GenTrain for validation; validation loss selects the checkpoint. Because language-balancing oversampling precedes the split, duplicated Chinese records may occur in both partitions.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). Under the same five-run protocol, the RCNN macro-averaged F1 scores computed from each run’s confusion matrix are 0.6678 ± 0.036 for sarcasm and 0.9298 ± 0.005 for MGTD. Sarcasm macro-F1 is higher than its positive-class F1 (0.6309) because the non-sarcastic majority class has a higher F1. These values supplement the positive-class results in §5.1; because macro-F1 is not reported for every configuration, they are not used for cross-configuration ranking.

C.2 Complete four-quadrant results for the MGTD head (including MGTD-only controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual-head	91.4	91.5	3.6	6.1
Interaction-projection dual-head	85.4	85.4	1.8	4.1
RCNN dual-head	94.8	93.2	5.5	6.6
Plain MGTD-only	90.4	86.6	4.1	4.2
RCNN MGTD-only	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

In every subgroup, each MGTD-only control differs from its corresponding dual-head model by fewer than ten percentage points. No comparably large subgroup error appears on the MGTD side, consistent with §5.3.

When the same detectors evaluate the Qwen3.5-generated AI test subset without retraining, the MGTD head again shows no comparable subgroup increase: AI × non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for the plain, interaction-projection, and RCNN configurations; AI × sarcastic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.965.

C.3 Evaluation protocol, inter-judge rank correlations, and group-level tests (§6). Each sample is evaluated separately by each of the three judge APIs in one API call using the same prompt structure for all model conditions (May 2026; temperature = 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated

text, not the ground-truth source label. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. The ratings are therefore obtained without ground-truth source labels, although the judges are explicitly asked to infer source before assigning the two ratings. Base and SFT outputs are evaluated separately.

Pairwise inter-judge rank correlations are low to moderate, with Spearman ρ ranging from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). We therefore do not treat individual-text rankings as a common gold standard. The condition-level observation rests separately on the consistent rating direction across all six language × judge groups.

Outputs from the SFT model receive lower mean sarcasm ratings in all six language × judge groups, and Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that five differences remain significant. The pooled correlations in Table 8 use the three-judge mean as a descriptive aggregate, while the limited pairwise rank correlations make judge-specific results necessary. Four instances each have one unparsable rating but retain two valid judge ratings, so averaging the available ratings preserves all 600 outputs in the aggregate.

Subset	n	Sarc. ρ	Human-like ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 8: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT): Sarc. ρ compares $P(\text{sarcastic})$ with sarcasm ratings, and Human-like ρ compares $P(\text{human})$ with human-likeness ratings. * significant; the other entries are not significant.

C.4 Paired tests between SFT outputs and human-authored reference texts (supplementary). These tests are performed on the Chinese subset of the 300-prompt generation evaluation ($n = 180$; three GPT-5-nano ratings are missing, leaving 177). The comparisons provide judge- and dimension-specific comparison context: for DeepSeek-V3, SFT outputs receive significantly lower human-likeness ratings than the reference texts (−0.161, $p = 0.0025$), whereas for GPT-5-nano, SFT outputs receive significantly higher

Judge / lang.	<i>n</i>	Base	SFT	adj. <i>p</i>
DeepSeek-V3 / en	120	4.025	3.292	<.0001
DeepSeek-V3 / zh	180	4.044	3.878	.0008
Gemini-2.5-FL / en	120	3.992	3.750	.0008
Gemini-2.5-FL / zh	180	4.028	3.994	.2008*
GPT-5-nano / en	119	4.084	3.681	<.0001
GPT-5-nano / zh	180	3.911	3.517	<.0001

Table 9: Group-level tests for sarcasm ratings obtained without ground-truth source labels (Wilcoxon + BH-FDR). * not significant. GPT-5-nano lacks one English base rating (C.3: $n = 119$) and three Chinese reference ratings (affecting C.4, not the C.3 Chinese row); tests use pairwise deletion.

sarcasm ratings than the reference texts (+0.186, $p = 0.0114$). The remaining four comparisons do not reach significance at the reported threshold. Thus, the observed reference differences vary across judges and dimensions; these tests do not estimate a judge-by-dimension interaction.

C.5 Plain-detector control. With the plain detector, the Chinese sarcasm probability changes from 0.765 to 0.763 (-0.001), the English sarcasm probability changes from 0.843 to 0.659 (-0.184), and predicted human-authorship probability rises by +0.760 in Chinese and +0.705 in English. The two detectors agree on the rise in predicted human-authorship probability. The Chinese sarcasm-probability change is detector-sensitive, whereas both detectors show a decrease in English.

Overall correlations between detector probabilities and corresponding ratings are +0.142 for sarcasm and -0.214 for human-likeness. The corresponding values are -0.037 (not significant) and -0.209 in Chinese, and +0.342 and -0.283 in English. Each differs from the RCNN-detector value by at most 0.02. Negative associations between predicted human-authorship probabilities and human-likeness ratings are similar across the two detectors, whereas the condition-level sarcasm-probability change differs between the plain and RCNN detectors, especially in Chinese.

C.6 Within-condition correlations (RCNN detector). For predicted human-authorship probability and human-likeness ratings, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese +0.10, English -0.09). The SFT condition has $\rho = -0.12$ ($p = 0.039$; Chinese -0.19 , $p = 0.009$; English -0.12 , not sig-

nificant). The pooled value of -0.212 is driven mainly by the opposing condition-level changes: SFT raises predicted human-authorship probability by about 0.67 while lowering the mean rating. The pooled negative correlation should therefore not be interpreted as a strong instance-level inverse relationship. Within either condition, predicted human-authorship probability is not significantly positively correlated with rated human-likeness.

For sarcasm, within-condition correlations are +0.05 for Chinese base (not significant), +0.15 for Chinese SFT ($p = 0.048$), +0.41 for English base, and +0.37 for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.7 Detector probabilities before and after SFT across the two task–language settings. The RCNN detector assigns the following mean probabilities to base and SFT outputs:

Lang.	Cond.	$P(\text{sarcastic})$	$P(\text{human})$
Chinese	Base $\rightarrow$ SFT	0.780 $\rightarrow$ 0.895	0.289 $\rightarrow$ 0.969
English	Base $\rightarrow$ SFT	0.854 $\rightarrow$ 0.730	0.198 $\rightarrow$ 0.864

Table 10: Mean detector probabilities before and after SFT (RCNN detector).

Sarcasm probabilities move in opposite directions in the Chinese comment-generation and English rewriting settings (+0.115 and -0.124 , respectively), while predicted human-authorship probabilities rise sharply in both. GenTrain itself is balanced at 1,214 instances per language, but the evaluation settings differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast.

The mean LLM-judge ratings for the human-authored reference texts are 3.77 for sarcasm and 4.17 for human-likeness; the corresponding values are 4.01 and 4.24 for the base model and 3.71 and 4.08 for SFT. From the base model to SFT, the mean human-likeness rating decreases even as predicted human-authorship probability increases, as analyzed in §6.

C.8 Uncertainty from test instances: confidence intervals, paired tests, and subset F1. For each separately trained final model, we quantify uncertainty from the test instances with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap sam-

ples for F1. The instance-level bootstrap resamples text instances independently and does not preserve human–AI seed pairs. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

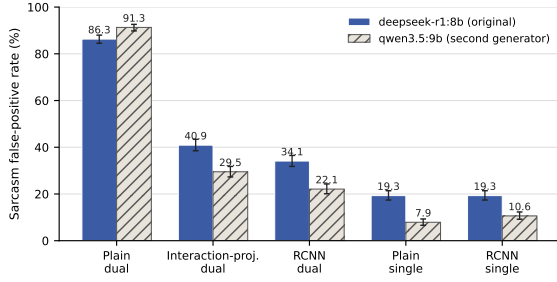


Figure 2: Sarcasm false-positive rates on AI-generated texts prompted to be non-sarcastic, for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

For both matched model pairs, the false-positive rate remains higher under the joint procedure on the Qwen3.5-generated texts, although the magnitudes differ.

Model	DeepSeek	Qwen3.5
Plain dual-head	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Interaction-projection dual-head	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual-head	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single-head	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single-head	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on AI-generated texts prompted to be non-sarcastic (two generators; %).

Paired McNemar tests use predictions on the same DeepSeek-generated AI $\times$ non-sarcastic texts. The discordant-pair counts are 1,021 / 7 for plain dual vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for the interaction-projection dual-head model vs. the RCNN dual-head model (all $p < .001$). The discordant counts and absolute rate differences provide the substantive effect-size information.

In both architectures and both language strata, the false-positive rate is higher under the joint

procedure than under its matched sarcasm-only control. Language remains coupled with corpus, genre, and annotation setting, so the stratified results do not isolate a language effect.

The final RCNN dual-head model’s MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the human-only subset contains no AI-authored examples, an MGTD F1 score on that subset would not measure source discrimination; Table 7 therefore reports human recall directly.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). This contrast also reflects differences in prevalence, genre, and annotation procedure: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values should therefore not be interpreted as an isolated language effect.

The separately retrained final RCNN model’s overall sarcasm F1 is 0.663, compared with 0.629 on human-authored texts alone, because the overall score also includes AI-generated texts. This point estimate differs from the five-run mean in Table 1, which aggregates fold-based runs. Within the human subset, absolute joint-versus-sarcasm-only differences in sarcasm F1 are below 0.03 and vary in direction. The negative-transfer effect is therefore concentrated in the quadrants containing AI-generated texts, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore lies mostly above the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

The median differences above show that joint training raises sarcasm scores on AI-generated texts, while the similar AUC values across models (0.82–0.88) show that their internal ordering is largely preserved. This pattern motivates the representation analyses in Appendix E.

Model	Lang.	n	Dual FP; rate [CI]	Single FP; rate [CI]	Δ pp
Plain	en	1,174	987; 84.1 [81.9, 86.1]	242; 20.6 [18.4, 23.0]	+63.5
Plain	zh	339	319; 94.1 [91.1, 96.1]	50; 14.7 [11.4, 18.9]	+79.4
RCNN	en	1,174	445; 37.9 [35.2, 40.7]	247; 21.0 [18.8, 23.5]	+16.9
RCNN	zh	339	71; 20.9 [17.0, 25.6]	45; 13.3 [10.1, 17.3]	+7.7

Table 12: Language-stratified sarcasm false-positive rates on AI-generated texts prompted to be non-sarcastic (%; Wilson 95% intervals). Dual FP and Single FP give the number of false positives under the dual- and single-head models; Δ is the dual rate minus the single rate. Exact paired McNemar tests give $p < .001$ for both plain comparisons and RCNN English, and $p = 2.4 \times 10^{-5}$ for RCNN Chinese.

Model	Full set	Chinese	English	Human subset	Human $\times$ zh	Human $\times$ en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 13: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for RCNN).

D Illustrative Cases of Disagreement between Detector Probabilities and LLM-Judge Ratings

Illustrative cases were manually selected after inspecting the paired table of RCNN-detector probabilities and ratings from the three LLM judges to show large disagreements; no formal ranking or top- k selection was performed. The selection is non-systematic, and the cases should not be used to estimate the prevalence of any disagreement type. Detector probabilities range from 0 to 1, and LLM-judge ratings are averaged across the three evaluators on a 1–5 scale. Long texts are shortened for display, while Chinese examples retain the original model input and provide Hanyu Pinyin transliterations and English translations in brackets.

The three groups show, respectively, high predicted human-authorship probabilities with low LLM-judge ratings of human-likeness, high predicted sarcasm probabilities with low LLM-judge ratings of sarcasm, and the reverse sarcasm disagreement. In the first human-likeness case, AI-generated text with an obvious reasoning-style preface is nevertheless assigned predicted human-authorship probability 1.00 by the MGTD head, corresponding to the disagreement discussed in §6.

Together, these cases motivate candidate hypotheses involving surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences, as well as semantic reversal, whose prevalence and causal effects require systematic testing. The examples illustrate several disagreement types compatible with the near-zero aggregate Chinese sarcasm correlation in §6.

Language	Model condition	Detector probability	Mean LLM-judge rating	Text (excerpt)
Chinese	SFT	1.00	2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [Hao, wo xuyao zhendui Sanxing huidui Huawei de xinwen shengcheng yi tiao fengci pinglun. Shouxian...; “Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (reasoning-style preface)
Chinese	SFT	1.00	2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [Paofen jingsai: canshu jiaolv: paofen jiaolv: shuihuo: hahaha; “Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
English	SFT	1.00	2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 14: Cases with high predicted human-authorship probability but low LLM-judge ratings of human-likeness. The detector column reports $P(\text{human})$.

Language	Model condition	Detector probability	Mean LLM-judge rating	Text (excerpt)
Chinese	SFT	1.00	2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sanxing: wo cai shi diyi! Ni kankan wo zhe shexiangtou zongfen, 109+96, 112+89, wo ying le!; “Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
Chinese	SFT	0.99	2.67	“恭喜恭喜，美国牛牛牛，继续加油” [Gongxi gongxi, Meiguo niu niu niu, jixu jiaoyou; “Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
English	SFT	0.93	2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 15: Cases with high predicted sarcasm probability but low LLM-judge ratings of sarcasm. The detector column reports $P(\text{sarcastic})$.

Language	Model condition	Detector probability	Mean LLM-judge rating	Text (excerpt)
Chinese	Base	0.14	4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [Telangpu chengnuo 120 yi butie xiang sa dabing, nongmin shoudao de que shi suoshui cheng yingbi de chengnuo...; “Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
Chinese	Base	0.24	4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [Tesla: jiage tiaoshui hou you tiaoshui, xiaofeizhe: qianbao tiaoshui hou you tiaoshui; “Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
English	Base	0.12	4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 16: Cases with low predicted sarcasm probability but high LLM-judge ratings of sarcasm. The detector column reports $P(\text{sarcastic})$.

E Representation-Level Analysis of Error Distribution between Source Groups

This appendix provides additional protocol details, diagnostic results, and implementation checks for the representation-level tests in §5.3.3.

E.1 Setup. The primary diagnostics use the frozen plain dual-head model that produces the 86.3% result reported in the main text; the matched sarcasm-only model provides the source-probe comparison. The full-set probe and logit-correlation analyses use all applicable DualTest examples. Intervention examples are divided with StratifiedShuffleSplit (test size 0.5, random state 42), using the four source $\times$ sarcasm groups as strata. The fitting/evaluation counts are 411/412 for human $\times$ sarcastic, 774/775 for human $\times$ non-sarcastic, 390/389 for AI $\times$ sarcastic, and 757/756 for AI $\times$ non-sarcastic. The fitting half contains 964 Chinese and 1,368 English texts; the evaluation half contains 934 Chinese and 1,398 English texts. Human-AI pair links are not used in the split, and 24.7% cross the two halves.

The trained task-head directions are read directly from the frozen model. The fitting half estimates probe, INLP, PCA, and erasure transformations, which are fixed and applied to the evaluation half. The original sarcasm head computes false-positive rates from transformed evaluation representations, while a new source probe is cross-validated within each transformed evaluation half to measure residual linear source decodability. The evaluation baseline is 84.788% for AI-generated non-sarcastic texts and 32.774% for human-authored non-sarcastic texts (gap 52.014 pp), with source AUC 0.98197.

Implementation checks show that manually computed logits match model outputs with a maximum error of 1.4×10^{-6} and that the residual after the erasure projection is about 10^{-6} . LEACE satisfies its defining property: after erasure, the difference between class means on the fitting half has a relative residual of 7×10^{-6} , and a linear probe evaluated on that half is at chance (AUC 0.514). This in-sample result serves only as an implementation check, whereas Table 17 reports probe performance after erasure on the held-out half. The AUC of the source probe is 0.986 with true labels, and a permuted-label sanity check gives 0.512. These probe values refer to different evaluation sets: 0.986 is the five-fold result on the unmodi-

fied full representation set, 0.982 is the corresponding baseline on the held-out half used to evaluate the interventions, 0.514 is the in-sample LEACE implementation check, and 0.589 is probe performance after erasure on the held-out half.

E.2 Protocol. Source probes use scikit-learn logistic regression with the `lbfgs` solver, ℓ_2 penalty, $C = 1.0$, no class weighting or feature scaling, and tolerance 10^{-4} . Single probes use 2,000 maximum iterations; each INLP step uses 1,500. AUC is computed by `cross_val_score` with five-fold stratified cross-validation, no shuffling, and ROC AUC scoring. Full-set cross-validated probes quantify baseline source decodability. A separate fitting-half probe supplies the direction used for alignment and single-direction intervention, while newly cross-validated probes within transformed evaluation representations quantify residual decodability. Permuted-label controls use `default_rng(0)`, and correlation analyses are repeated within text-length tertiles.

Directions and alignment. Each task head is Dropout followed by `Linear(1024,2)`. Its direction is the class-1 minus class-0 weight vector: class 1 denotes sarcasm for the sarcasm head and human authorship for MGTD, so the AI direction is the negative MGTD weight difference. Directional analyses ignore the bias, whereas predictions use the complete linear head with bias and dropout disabled in evaluation mode. The fitting-half source-probe direction is the ℓ_2 -normalized logistic-regression weight after subtracting the full-set pooled mean and is oriented toward human authorship.

Let Σ_{all} be the Ledoit–Wolf shrinkage covariance (Ledoit and Wolf, 2004) fitted to all pooled DualTest representations, with eigenvalues clipped below 10^{-8} . Covariance-aware alignment is $\cos(\Sigma_{\text{all}}^{1/2} w_1, \Sigma_{\text{all}}^{1/2} w_2)$. The signed values are -0.41197 for the fitting-half source-probe versus sarcasm-head directions and $+0.48713$ for the MGTD- versus sarcasm-head directions; the main text reports their absolute values because the sign depends on whether each source direction is oriented toward human or AI authorship. Because Σ_{all} uses all unlabeled DualTest representations, these alignment measurements are transductive.

Centered Euclidean erasure and INLP. For representations X , mean μ , and a matrix U with orthonormal direction columns, Euclidean erasure is

$$\mathcal{E}_\mu(X, U) = X - [(X - \mu)U]U^\top, \quad (3)$$

which retains the mean component. The task-head, single-probe, INLP, and Euclidean random-direction and random-subspace controls use the full-set pooled mean; LEACE, the covariance-whitened random controls, and srcORTH-PC1 use the fitting-half mean. INLP iteratively fits a source probe, normalizes its weight, orthogonalizes it against previously found directions by Gram–Schmidt, and deflates the residual representation. We evaluate $k \in \{1, 2, 4, 8, 16\}$ removed directions. Same-dimensional random-subspace controls orthogonalize Gaussian matrices by QR decomposition; both these controls and the permuted-label controls use `default_rng(0)`.

LEACE and covariance-whitened controls. LEACE is implemented directly rather than through an external concept-erasure package. On the fitting half, let Σ_{fit} be the Ledoit–Wolf covariance with eigenvalues clipped below 10^{-8} , and let $S_{xz} = X_c^\top Z_c / n$ for centered representations X_c and centered one-hot source labels Z_c . We compute $M = \Sigma_{\text{fit}}^{-1/2} S_{xz}$ and its singular vectors; the threshold $s > 10^{-6}$ gives rank one. With the retained whitened direction matrix Q , the oblique projection is $P = \Sigma_{\text{fit}}^{1/2} Q Q^\top \Sigma_{\text{fit}}^{-1/2}$, and erasure maps X to $X - (X - \mu_{\text{fit}})P^\top$. The ten covariance-whitened controls sample unit Gaussian directions q with `default_rng(0)` and use the corresponding rank-one oblique projection $\Sigma_{\text{fit}}^{1/2} q q^\top \Sigma_{\text{fit}}^{-1/2}$.

srcORTH-PC1. We fit an INLP source subspace S with three normalized, Gram–Schmidt-orthogonalized logistic-regression directions on the fitting half. With μ_{fit} as the fitting mean, the residual matrix is

$$R = (X_{\text{fit}} - \mu_{\text{fit}}) - [(X_{\text{fit}} - \mu_{\text{fit}})S]S^\top. \quad (4)$$

We compute `numpy.linalg.eigh` on `np.cov(R^\top)` and define srcORTH-PC1 as the eigenvector with the largest eigenvalue. It lies in the orthogonal complement of the fitted INLP-3 subspace. Evaluation removes srcORTH-PC1 alone from the original evaluation representations using Equation 3; the INLP-3 subspace is not removed first.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also yields an AUC of

0.974, showing that source decodability is a background property shared by both models. On $\text{AI} \times \text{non-sarcastic}$ texts, the MGTD-head AI logit and sarcasm-head logit correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and MGTD-head directions are nearly orthogonal in the original geometry ($\cos \approx -0.03$). Covariance-aware absolute cosine is 0.41 for the fitting-half source-probe direction and 0.49 for the MGTD-head direction. Together with the logit correlation, these measurements provide correlational evidence that source-related scores and directions are associated with the affected sarcasm decisions.

E.4 Intervention evidence. We define gap as the $\text{AI} \times \text{non-sarcastic}$ false-positive rate minus the human $\times$ non-sarcastic false-positive rate. Source AUC measures the performance of a probe re-trained after erasure.

Erasure scheme	AI $\times$ n FPR	H $\times$ n FPR	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
MGTD-head direction	86.2	30.2	56.0	0.982
Source-probe direction	84.8	32.9	51.9	0.981
Euclidean random direction	85.2	32.6	52.5	0.982
INLP ($k = 16$)	89.8	17.7	72.1	0.974
LEACE (linear)	68.7	49.5	19.1	0.589
Whitened random $\times 10$ (mean)	84.9	32.7	52.2	0.982
srcORTH-PC1	66.7	54.1	12.6	0.982

Table 17: Intervention summary. H $\times$ n = human $\times$ non-sarcastic; AI $\times$ n = AI $\times$ non-sarcastic; FPR in %; gaps are computed from unrounded FPRs. Values for the covariance-whitened random controls are means over ten sampled directions; their source-group gaps range from 51.4 to 52.7 pp. AUC is the AUC of the source probe after erasure. Bold marks the two focal interventions discussed in §5.3.3.

The MGTD-head, source-probe, Euclidean-random, and INLP interventions do not materially narrow the source-group gap. LEACE reduces measured linear source decodability and the gap defined above from 52.0 to 19.1. Its gap change is not matched by any of the ten directions sampled in covariance-whitened space.

The srcORTH-PC1 intervention changes the AI-generated subgroup rate to 66.7 and the difference between source groups to 12.6 while measured source decodability remains 0.982 at the reported precision. Thus, the distribution of errors between

source groups is sensitive to a high-variance direction in the orthogonal complement of the fitted INLP-3 source subspace.

We use an unweighted group mean to give the two non-sarcastic source groups equal importance despite their different sizes. Under LEACE, this mean false-positive rate is almost unchanged (58.8 $\rightarrow$ 59.1), confirming that LEACE reallocates non-sarcastic errors between AI and human texts.